

SEMICONDUCTOR SUPPLY CHAIN PERFORMANCE ANALYSIS

PROBLEM STATEMENT: *Why is our supply chain cost high, and which suppliers are responsible for it?*

End-to-End Analytics Project | Excel • Power Query • Power BI • MySQL

January – July 2025 | 200 Orders | 5 Suppliers | 5 Locations | 3 Product Categories

₹29.36 L	₹23.47 L	200	28.7 days	3.09%
Total Revenue	Procurement Cost	Total Orders	Average Lead Time	Overall Defect Rate

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1. Executive Summary

This report presents a comprehensive analysis of the semiconductor supply chain spanning January to July 2025, covering 200 procurement orders across 5 suppliers, 5 manufacturing locations, and 3 product categories. The project was executed end-to-end using Microsoft Excel and Power Query for data cleaning, Power BI for interactive dashboards, and MySQL for structured querying — replicating a real-world analytics workflow adopted by MNC supply chain and operations teams.

Core Business Question: Why is our supply chain cost high, and which suppliers are responsible for it?

The analysis reveals that procurement costs are concentrated in two suppliers (S4 and S5), that microprocessors account for 60% of all revenue but carry the highest unit cost, and that a 43.5% order failure rate in quality inspection is creating hidden rework and delay costs. Mumbai emerges as the highest-revenue hub but also the highest-cost location, and Bengaluru offers the best cost-to-revenue efficiency. No strong correlation was found between higher spending per supplier and better delivery or defect performance — meaning the business is not getting proportional value from its top-cost vendors.

Dimension	Key Finding
Cost Driver	S4 & S5 account for 57% of total procurement cost
Revenue Leader	S5 generates the highest revenue at ₹9.27 L (31.5% share)
Quality Risk	43.5% orders Failed inspection; S5 has the highest defect rate at 3.23%
Lead Time Alert	S4 has the slowest average lead time at 29.6 days
Best Efficiency	S3 — lowest lead time (26.8 days) + lowest defect rate (2.90%)
Top Location	Mumbai — highest revenue (₹8.6 L); Bengaluru — best cost:revenue ratio
Product Mix	Microprocessors: 60% revenue; Logic ICs: only 9% — portfolio imbalance

2. Project Overview & Objectives

2.1 Business Context

Semiconductor supply chains are among the most complex in modern manufacturing. Every rupee overspent on procurement, every delayed shipment, and every failed quality inspection translates to lost margins, production stoppages, and customer dissatisfaction. This project simulates the role of a Supply Chain Data Analyst at a mid-to-large semiconductor firm, tasked with answering four strategic questions for senior leadership:

- Why are procurement costs increasing, and how can supplier performance be optimized?
- Which suppliers are helping us grow, and which are hurting efficiency?
- Where are delays and quality issues occurring in our supply chain?
- Are we spending more money on suppliers who deliver better performance?

2.2 Dataset Overview

Attribute	Detail	Attribute	Detail
Source Format	Excel (.xlsx)	Total Records	200 orders
Time Period	Jan – Jul 2025	Suppliers	S1, S2, S3, S4, S5
Locations	5 Indian cities	Product Types	3 categories
Key Columns	15 fields	Tool Stack	Excel, Power Query, Power BI, MySQL

2.3 Column Glossary

The dataset contains 15 columns representing the full order-to-inspection lifecycle:

Column Name	Description	Business Use
Order_ID	Unique identifier per order	Primary key for order-level analysis
Supplier	Supplier code (S1–S5)	Vendor performance grouping
SKU	Stock Keeping Unit identifier	Product diversity & coverage
Product_Type	Category (Microprocessors, Sensors, Logic ICs)	Revenue mix analysis
Ordered_Quantity	Units ordered per transaction	Volume & procurement cost
Unit_Cost	Cost per unit (₹)	Procurement cost calculation
Order_Date	Date of order placement	Trend & seasonality analysis

Supplier_Lead_Time_Days	Days from order to supplier dispatch	Supplier speed benchmarking
Manufacturing_Lead_Time_Days	Days from dispatch to production ready	Production planning
Avg_Lead_Time	Average of both lead time components	Overall supply chain speed
Inspection_Result	Pass / Fail / Pending	Quality gate performance
Defect_Rate	% defective units in shipment	Quality risk scoring
Location	Destination manufacturing city	Geographic cost & revenue map
Revenue	Revenue generated per order (₹)	Profitability analysis
Total_Cost	Total procurement cost per order (₹)	Cost-to-revenue ratio

3. The Story — Why Is Our Supply Chain Cost High?

A Narrative Walk Through the Data

Act I — The Cost Accumulation Problem

Imagine you are a Supply Chain Director reviewing the books for the first half of 2025. Total procurement cost has hit ₹23.47 lakhs against revenue of ₹29.36 lakhs — a 79.9% cost-to-revenue ratio. The question your CFO is asking: where is the money going, and can we do better?

The answer lies in two suppliers. S4 and S5 together consumed ₹13.48 lakhs in procurement cost — 57% of the entire spend. S5 alone accounted for ₹7.44 lakhs. These aren't small vendors; they are the backbone of the supply chain. But are they earning that position?

Finding: S4 and S5 carry the highest costs AND the highest defect rates (3.22% and 3.23% respectively). High spend is not translating into high quality.

Act II — Which Suppliers Are Helping vs. Hurting?

When we rank suppliers across three dimensions — revenue generated, defect rate, and lead time — a clear picture emerges. S5 leads in revenue at ₹9.27 lakhs, but it also posts the worst defect rate in the portfolio. S3, by contrast, is the quiet outperformer: it has the lowest average lead time (26.8 days) and the lowest defect rate (2.90%), yet it receives the second-lowest procurement budget.

Supplier	Revenue (₹)	Proc. Cost (₹)	Avg Lead Time	Defect Rate	Assessment
S1	3,07,804	2,43,984	28.71 days	2.91%	Neutral
S2	5,77,202	4,62,198	28.58 days	3.20%	Watch
S3	3,65,700	2,91,814	26.81 days	2.90%	Best Performer
S4	7,58,891	6,05,039	29.58 days	3.22%	High Risk
S5	9,26,804	7,43,580	29.24 days	3.23%	High Risk

Act III — Where Are the Delays and Quality Failures?

Of 200 orders, only 72 (36%) passed quality inspection. 87 orders (43.5%) failed outright, while 41 (20.5%) remain pending — a troubling backlog. This is not a marginal quality issue; it is a systemic failure affecting nearly half of all shipments.

Geographically, Coimbatore bears the highest volume of orders but Mumbai generates the most revenue (₹8.6 lakhs). The Bengaluru hub, while smaller, delivers the best cost efficiency — ₹3.01 lakhs in cost for ₹3.74 lakhs in revenue. This is the model other locations should aspire to.

Quality Alert: 87 of 200 orders (43.5%) failed inspection. Microprocessors had the highest defect count. If even 20% of these require rework or replacement, the hidden cost could exceed ₹2 lakhs.

Act IV — Are We Paying for Performance?

The final question is the most strategically important: does spending more on a supplier mean getting better outcomes? The data says no. S4 and S5 command the top two positions in procurement spend but rank last in both lead time and defect rate. S3 — the best performer on both quality and speed — receives among the lowest investment. This misalignment is the core opportunity for the business: rebalancing the supplier portfolio to reward performance.

Strategic Insight: There is zero positive correlation between procurement spend per supplier and quality or speed performance. The organization is rewarding volume, not value.

4. Data Collection & Excel Processing

4.1 Raw Data Structure

The raw dataset was received as a flat Excel file (.xlsx) with 200 rows and 15 columns, representing order-level records for the semiconductor supply chain. Prior to any analysis, the data required profiling, cleaning, and structural validation to ensure analytical integrity.

4.2 Excel Cleaning Steps Performed

Step 1 — Data Profiling & Initial Review

- Opened the workbook and applied Freeze Panes on Row 1 to keep headers visible during scroll
- Used Ctrl+End to identify the true data range and confirm 200 data rows (Row 2 to Row 201)
- Applied AutoFilter to all columns for quick slice-and-dice review of unique values
- Checked for completely blank rows using Go To Special > Blanks — none found

Step 2 — Removing Duplicates

- Selected the full data range and applied Data > Remove Duplicates on Order_ID column
- Result: 0 duplicates found — dataset confirmed as unique at order level
- Created a backup sheet 'Raw_Data_Backup' before any transformations

Step 3 — Data Type Standardization

- Order_Date column: selected column and applied Format Cells > Date (YYYY-MM-DD) format to ensure consistent parsing
- Unit_Cost, Revenue, Total_Cost: formatted as Number with 2 decimal places
- Defect_Rate: formatted as Number (percentage was stored as decimal — e.g., 2.46 not 0.0246)
- Ordered_Quantity: confirmed as Integer — no decimal values present

Step 4 — Supplier & Location Validation

- Used COUNTIF formula to verify only S1–S5 exist in Supplier column: =COUNTIF(B:B, "S*") = 200
- Used Data Validation > List on Location column to confirm only 5 valid city names
- Used TRIM() function on text columns (Supplier, Location, Product_Type) to remove accidental leading/trailing spaces

Step 5 — Derived Column Verification

- Verified Total_Cost = Ordered_Quantity × Unit_Cost using a check column: =ABS(O2 - (E2*F2)) — all values returned 0 (within rounding)
- Verified Avg_Lead_Time = (Supplier_Lead_Time_Days + Manufacturing_Lead_Time_Days) / 2 for a sample of 20 rows
- Added a helper column 'Month_Year' using =TEXT(G2, "MMM-YYYY") for time-series grouping in pivot tables

Step 6 — Pivot Table Summaries for Validation

- Created Supplier Summary Pivot: Rows = Supplier | Values = SUM(Revenue), SUM(Total_Cost), AVERAGE(Defect_Rate), AVERAGE(Avg_Lead_Time)
- Created Monthly Trend Pivot: Rows = Month_Year | Values = SUM(Revenue), COUNT(Order_ID)
- Created Inspection Pivot: Rows = Inspection_Result | Values = COUNT(Order_ID) — confirmed 72 Pass, 87 Fail, 41 Pending
- Cross-validated all pivot totals against manual SUM formulas — totals matched

5. Power Query — Data Cleaning & Transformation

5.1 Why Power Query?

While Excel's manual cleaning steps validated data quality, Power Query (accessed via Data > Get & Transform > From Table/Range) was used to build a repeatable, auditable transformation pipeline. This mirrors the ETL (Extract, Transform, Load) workflows used in enterprise BI tools like Azure Data Factory and SSIS.

5.2 Power Query Steps — Applied Steps Panel

Step 1 — Source & Promote Headers

- Connected to the Excel table as source using 'From Table/Range'
- Applied 'Use First Row as Headers' to promote column names from Row 1
- Verified all 15 column names were correctly imported with no encoding issues

Step 2 — Data Type Assignment

- Order_ID: changed to Whole Number
- Order_Date: changed to Date type (Power Query auto-detected but required explicit confirmation)
- Unit_Cost, Revenue, Total_Cost, Defect_Rate: changed to Decimal Number
- Ordered_Quantity, Supplier_Lead_Time_Days, Manufacturing_Lead_Time_Days, Avg_Lead_Time: changed to Whole Number
- Supplier, SKU, Product_Type, Inspection_Result, Location: changed to Text

Step 3 — Text Normalization

- Applied Transform > Format > Trim to Supplier, Location, Product_Type, Inspection_Result columns to remove whitespace
- Applied Transform > Format > Clean to remove non-printable characters from text columns
- Applied Transform > Format > Capitalize Each Word to Location for display consistency

Step 4 — Null & Error Handling

- Used 'Keep Rows > Keep Non-Blank Rows' — no nulls found in primary ID and cost columns
- Added conditional column to flag rows where Inspection_Result was neither Pass, Fail, nor Pending — result: 0 anomalies
- Used 'Replace Errors' on numeric columns with 0 as fallback (no errors were present)

Step 5 — Derived Columns Added in Power Query

- Procurement_Cost: = [Ordered_Quantity] * [Unit_Cost] — calculated column for independent verification
- Month_Name: = Date.MonthName([Order_Date]) — for monthly grouping
- Month_Year: = Text.From(Date.Year([Order_Date])) & "-" & Text.PadStart(Text.From(Date.Month([Order_Date])),2,"0") — for sort-friendly month labels
- Gross_Margin: = [Revenue] - [Total_Cost] — for profitability layer
- Lead_Time_Band: conditional column — <27 days = 'Fast', 27–30 = 'Average', >30 = 'Slow'
- Quality_Flag: conditional column — Defect_Rate > 3.0 = 'High Risk', else 'Acceptable'

Step 6 — Load & Refresh Settings

- Loaded the cleaned query to the Data Model (Power Pivot) to enable cross-table DAX measures
- Set refresh on file open to ensure live data updates propagate automatically to Power BI
- Named the query 'SCM_Cleaned' for clear identification in the queries panel

5.3 DAX Measures Created (for Power BI)

The following DAX measures were defined in Power BI Desktop after connecting to the cleaned dataset:

Measure Name	DAX Formula
Total Revenue	Total Revenue = SUM(SCM_Cleaned[Revenue])
Total Procurement Cost	Total Proc Cost = SUM(SCM_Cleaned[Total_Cost])
Avg Defect Rate	Avg Defect Rate = AVERAGE(SCM_Cleaned[Defect_Rate])
Avg Lead Time	Avg Lead Time = AVERAGE(SCM_Cleaned[Avg_Lead_Time])
Cost to Revenue Ratio	Cost:Rev Ratio = DIVIDE([Total Proc Cost],[Total Revenue],0)
Pass Rate %	Pass Rate % = DIVIDE(COUNTROWS(FILTER(SCM_Cleaned, SCM_Cleaned[Inspection_Result]="Pass")), COUNTROWS(SCM_Cleaned), 0)
Fail Rate %	Fail Rate % = DIVIDE(COUNTROWS(FILTER(SCM_Cleaned, SCM_Cleaned[Inspection_Result]="Fail")), COUNTROWS(SCM_Cleaned), 0)
Revenue Growth MoM	Rev Growth = DIVIDE([Total Revenue] - CALCULATE([Total Revenue], DATEADD(SCM_Cleaned[Order_Date],-1,MONTH)), CALCULATE([Total Revenue], DATEADD(SCM_Cleaned[Order_Date],-1,MONTH)), 0)
Supplier Rank (Rev)	Supplier Rank = RANKX(ALL(SCM_Cleaned[Supplier]),[Total Revenue],,DESC)

6. Power BI Dashboard — Visuals, Insights & Outcomes

6.1 Dashboard Architecture

The Power BI report was structured across three pages to support different stakeholder audiences — a high-level executive summary, a supplier deep-dive, and a location/product breakdown. All pages use slicers for Supplier, Location, Product_Type, and Order_Date (date range) to enable self-service filtering.

6.2 Page 1 — Executive Overview

Visuals Included

- KPI Cards (5): Total Revenue | Total Procurement Cost | Total Orders | Avg Lead Time | Avg Defect Rate
- Clustered Bar Chart — Revenue vs. Procurement Cost by Supplier (side-by-side comparison)
- Donut Chart — Inspection Result Distribution (Pass / Fail / Pending with % labels)
- Line Chart — Monthly Revenue Trend (Jan–Jul 2025 with data labels)
- Treemap — Revenue by Product Type (area proportional to revenue share)

Key Insights Surfaced on Page 1

- S5 and S4 together generate ₹16.86 lakhs in revenue but consume ₹13.48 lakhs in cost — a 80% cost burden ratio
- The donut chart immediately reveals the inspection crisis: Fail (43.5%) > Pass (36%) — a visual shock for any executive audience
- The monthly trend line shows a sharp drop in June–July 2025 (₹5.03 L → ₹2.46 L), signaling a potential demand slowdown or supply disruption that requires investigation
- Microprocessors occupy 80%+ of the treemap, confirming product concentration risk

6.3 Page 2 — Supplier Performance Scorecard

Visuals Included

- Matrix Table — Supplier × [Revenue, Cost, Avg Lead Time, Defect Rate, SKU Count] with conditional formatting (green-to-red gradient on Defect Rate)
- Scatter Plot — Avg Lead Time (X) vs. Avg Defect Rate (Y), bubble size = Revenue — identifies the 'ideal quadrant' (low-low)
- 100% Stacked Bar — Supplier contribution to total procurement cost (% breakdown)
- Bar Chart — SKU count per supplier (product breadth comparison)
- Waterfall Chart — Cumulative revenue build-up across suppliers in rank order

Key Insights Surfaced on Page 2

- Scatter plot places S3 in the ideal bottom-left quadrant (fast + low defect) while S4 and S5 cluster in the top-right (slow + high defect) — a compelling visual for supplier renegotiation
- S3 has 26 unique SKUs vs. S5's 42 — but S3 delivers better quality per SKU, suggesting product focus over breadth

- S5 contributes 31.7% of total procurement cost yet has the highest defect rate — the cost-quality misalignment is visible at a glance
- S1 is the most 'neutral' supplier — mid-range on all dimensions, offering no standout risk or value

6.4 Page 3 — Location & Product Analysis

Visuals Included

- Map Visual (Filled) — Indian map with city bubbles sized by Revenue and colored by Defect Rate
- Grouped Bar Chart — Revenue vs. Cost by Location (5 cities side-by-side)
- Stacked Column Chart — Revenue by Product_Type within each Location
- Table Visual — Location × [Revenue, Cost, Cost:Revenue Ratio, Fail Order Count] with data bars
- Slicer Panel — Month selector, Supplier multi-select, Product_Type checkbox

Key Insights Surfaced on Page 3

- Mumbai leads on revenue (₹8.60 L) but also has the highest absolute cost (₹6.82 L) — the margin is thinner than it appears
- Bengaluru has a cost:revenue ratio of 0.807 — the best in the portfolio — making it the benchmark for operational efficiency
- Coimbatore and Chennai together handle 46% of orders but generate only 39% of revenue — a volume-without-value problem
- Logic ICs generate almost no significant revenue in any location, raising the question of whether this product line is strategically viable
- Pune shows the highest growth trajectory in the May–July window — a location worth investing in

6.5 Cross-Page Insights — Answering the Four Business Questions

Business Question	Power BI Answer
Why are procurement costs increasing?	S4 and S5 account for 57% of all cost. Microprocessors are the highest unit-cost category. No volume discount effect is visible — cost scales linearly with quantity.
Which suppliers are helping vs. hurting?	S3 = best performer (fast, low defect). S5 = highest revenue but worst quality. S4 = highest cost per unit and slowest lead time.
Where are delays and quality issues?	S4 drives lead time delays. 43.5% fail rate is systemic across all suppliers but worst for S4 and S5. Microprocessors have the highest fail count by product type.
Are we paying for performance?	No. The scatter plot (Page 2) shows zero correlation between procurement spend and quality/speed outcomes. S3 delivers best value at below-average cost.

7. SQL Analysis — Query Breakdown & Key Findings

7.1 Database Setup

The cleaned dataset was exported from Power Query as a CSV and imported into a MySQL 8.0 database. The table was named 'scm_table' and indexed on Order_ID, Supplier, and Order_Date for query performance. All 35 queries were written and tested in MySQL Workbench 8.0.

7.2 Query Categories & Business Mapping

Q#	Category & Query	Business Purpose & Finding
Q1	Total orders placed	Baseline: 200 orders confirmed across the period
Q2	Total procurement cost	₹23,46,615 total cost — the number the CFO wants to reduce
Q3	Supplier with highest procurement cost	S5 — ₹7,43,580 — largest single cost driver in the portfolio
Q4	Average cost per unit by supplier	S4 has highest avg unit cost; S1 has lowest — pricing strategy gap
Q5	Location with highest procurement cost	Mumbai — ₹6,81,558 — needs cost efficiency review
Q6	Total quantity by supplier	S1 ordered most units (2,459) — high volume, low unit cost
Q7	Supplier with maximum quantity supplied	S1 — volume leader despite not being cost leader
Q8	SKU count per supplier	S1 covers 56 unique SKUs — broadest product coverage
Q9	Supplier with widest SKU coverage	S1 — most versatile supplier; S3 narrowest (26 SKUs)
Q10	Average lead time per supplier	S3 fastest (26.8 days); S4 slowest (29.6 days)
Q11	Supplier with lowest avg lead time	S3 — operational efficiency leader in speed
Q12	Month-wise average lead time	Lead time ranged from 27.4 to 30.1 days across months
Q13	Month with highest avg lead time	Identified the bottleneck month for targeted investigation
Q14	Overall defect rate	3.09% average — above acceptable 2% industry threshold
Q15	Defect rate per supplier	S5 worst (3.23%), S3 best (2.90%) — clear quality stratification
Q16	Inspection result count (Pass/Fail/Pending)	Fail: 87 Pass: 72 Pending: 41 — critical quality alert
Q17	Product type with highest defect count	Microprocessors — highest fail count, highest risk category

Q18	Total revenue	₹29,36,400 — baseline for all profitability analysis
Q19	Month-wise revenue trend	March peak (₹5.42 L); July lowest (₹2.46 L) — sharp deceleration
Q20	Month with highest revenue	March 2025 — peak procurement and production activity
Q21	Revenue by product type	Microprocessors ₹17.72 L (60%); Logic ICs only ₹2.65 L (9%)
Q22	Supplier with highest revenue	S5 — ₹9.27 L (31.5%) — revenue anchor of the business
Q23	Average revenue per order	₹14,682 per order — useful for order value benchmarking
Q24	Revenue by location × product type	Mumbai+Microprocessors is the single largest revenue combination
Q25	Location with best cost-to-revenue ratio	Bengaluru — ratio 0.807 — most efficient city in the network
Q26	% contribution to total procurement cost	S5: 31.7% S4: 25.8% S2: 19.7% S3: 12.4% S1: 10.4%
Q27	Rank suppliers by procurement cost	S5 > S4 > S2 > S3 > S1 — cost concentration in top 2 is clear
Q28	Rank suppliers by avg lead time	S3 fastest (rank 1) S4 slowest (rank 5) — performance inverse of cost rank
Q29	Suppliers with above-avg defect rates	S2, S4, S5 all above 3.09% average — three suppliers need quality contracts
Q30	Cumulative revenue over time	Revenue milestone tracking — reached ₹14.7 L by end of March
Q31	Top 3 suppliers by revenue (window fn)	S5 (1st, ₹9.27 L), S4 (2nd, ₹7.59 L), S2 (3rd, ₹5.77 L)
Q32	Top supplier per product category	Microprocessors: S5 Sensors: S4 Logic ICs: S3
Q33	Supplier × location: highest cost combo	S5 + Mumbai is the single costliest supply chain node
Q34	Suppliers with low cost AND low lead time	S3 qualifies — the ideal strategic partner for volume growth
Q35	Flag: high cost + high defect rate	S4 and S5 both flagged — priority targets for renegotiation or replacement

7.3 Highlighted SQL Patterns Demonstrated

Pattern 1 — Aggregation & Grouping (Q2–Q9, Q14–Q18)

Standard GROUP BY with SUM, AVG, COUNT, and DISTINCT was used throughout procurement and quality analysis. Key skill: combining multiple aggregations in a single query to produce multi-metric supplier scorecards without multiple round trips.

Pattern 2 — Subquery Thresholding (Q29, Q34, Q35)

HAVING clauses with scalar subqueries were used to dynamically benchmark suppliers against the dataset average — a technique used in real-world supplier tiering systems. This avoids hard-coding thresholds and makes the logic self-updating when data refreshes.

Example (Q35): `HAVING AVG(Unit_Cost) > (SELECT AVG(Unit_Cost) FROM scm_table)`
`AND AVG(Defect_Rate) > (SELECT AVG(Defect_Rate) FROM scm_table)`

Pattern 3 — Window Functions (Q26–Q32)

Advanced analytics using `RANK()`, `DENSE_RANK()`, `SUM() OVER`, and `RANKX`-equivalent logic in MySQL 8.0. These are the patterns that separate junior SQL writers from analysts ready for enterprise reporting.

Example (Q26): `SUM(cost) * 100.0 / SUM(SUM(cost)) OVER ()` — calculates % share without a self-join

Pattern 4 — Date Functions & Time Series (Q12, Q13, Q19, Q20, Q30)

`DATE_FORMAT()` for month-year grouping and cumulative `SUM OVER (ORDER BY Order_Date ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW)` for running totals — patterns directly used in financial reporting pipelines.

Pattern 5 — PARTITION BY for Category Leaders (Q32)

`RANK() OVER (PARTITION BY Product_Type ORDER BY SUM(Revenue) DESC)` identifies the top supplier within each product category independently — a common requirement in sales and supply chain dashboards.

8. Strategic Recommendations

8.1 Procurement Cost Reduction

- Renegotiate contracts with S4 and S5 — current spend concentration (57% of procurement) with poor quality outcomes justifies price or SLA renegotiation
- Pilot volume reallocation: shift 15–20% of orders from S4/S5 to S3, which has proven lower lead time and defect rates
- Implement unit cost benchmarking quarterly; S4's avg unit cost is the highest in the portfolio with no quality premium to justify it

8.2 Quality Improvement

- Set a maximum defect rate threshold of 3.0% as a contract SLA for all suppliers — S2, S4, and S5 currently exceed this
- Require pre-shipment inspection certificates for all Microprocessor orders — the category with the highest fail count
- Move Pending inspections to a 7-day resolution SLA — 41 open inspections (20.5% of orders) represent undetermined risk
- Create a Supplier Quality Scorecard updated monthly, visible to procurement and operations leadership

8.3 Lead Time Optimization

- Use S3 as the benchmark: 26.8 days average. Target S4 (currently 29.6 days) to reduce lead time by at least 2 days within Q3
- Introduce safety stock for Microprocessors at Mumbai and Coimbatore locations given their high volume + high lead time combination
- Evaluate whether Manufacturing Lead Time for high-volume SKUs can be parallelized to reduce the Avg_Lead_Time dependency on sequential processes

8.4 Supplier Portfolio Strategy

- S3 is the portfolio's best-kept secret — expand its share of business, particularly for Sensors and Logic ICs where it is already the top revenue generator
- S5 should be placed on a performance improvement plan: revenue leadership cannot co-exist with highest defect rate indefinitely
- Bengaluru operations should be studied and replicated — its cost:revenue efficiency is the standard to match across all 5 locations
- Evaluate Logic ICs viability: at 9% revenue share with no location showing significant traction, a strategic review of this product line is warranted

9. Conclusion

This project delivered a full-stack supply chain analytics solution — from raw Excel data through Power Query cleaning, Power BI visualization, and MySQL querying — answering four strategic questions that matter to C-suite leadership in any manufacturing or semiconductor firm.

The central finding is clear: procurement costs are high because spend is concentrated in underperforming suppliers. S4 and S5 consume over half of the procurement budget while posting the worst quality and delivery metrics in the portfolio. S3 — underfunded and overlooked — is the operational benchmark the business should build toward.

The 43.5% inspection failure rate is the most urgent operational risk identified. Unresolved, it will continue to generate hidden rework costs, delay costs, and customer dissatisfaction that no amount of procurement renegotiation can fully offset.

The recommendations in Section 8, if implemented, project a potential 10–15% reduction in procurement cost and a 20–30% improvement in inspection pass rates within two quarters — based on realigning spend toward proven suppliers and enforcing quality SLAs.

10. Skills & Tools Demonstrated

Tool / Skill	Techniques Applied
Microsoft Excel	Data profiling, duplicate removal, data type formatting, TRIM/COUNTIF/TEXT formulas, Pivot Tables, cross-validation, conditional formatting
Power Query	ETL pipeline design, type assignment, text normalization, null handling, derived columns (calculated fields, conditional columns), data model load
DAX (Power BI)	SUM, AVERAGE, COUNTROWS, FILTER, DIVIDE, CALCULATE, DATEADD, RANKX, ALL — 9 production-grade measures
Power BI Desktop	3-page dashboard, 15+ visual types, KPI cards, scatter plots, treemaps, matrix with conditional formatting, cross-page slicers, map visual
MySQL 8.0	GROUP BY aggregation, HAVING with subqueries, DATE_FORMAT, RANK/DENSE_RANK, SUM OVER, PARTITION BY, cumulative window functions, multi-condition filtering
Analytical Thinking	Business question framing, insight narrative, supplier benchmarking, cost-quality correlation analysis, strategic recommendation development
Report Writing	MNC-standard structured report, executive summary, data storytelling, table-based evidence, actionable recommendations

— END OF REPORT —